GGS 662 AGENTIC GEOAI · RESEARCH PROGRESS

Seeing Through Clouds

Handling clouds in agentic change detection

TEST CASE: HURRICANE IDA 2021 · SPACENET 8 · 0.5 M EO

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Preliminary results on development data · Imagery: Maxar via SpaceNet 8



Clouds hide what our change detector must see

The system. A vision foundation model finds where two images differ, an agent describes the change, and code checks every fact. For now, flood analysis: each road's features go into our model, which predicts flooded or not.

ONE DIFFICULTY · CLOUDS IN EO IMAGERY

"Clouds can hide target areas in images taken after disasters. This can lead to labels being inaccurate compared to ground truth due to obscured parts of the imagery. How should we detect clouds in this imagery, handle them, and report them? The goal is results, such as maps of flooded roads, that people can trust."

Expected contribution: a tested rule for what a change map may say under cloud, a measured label gap, and a test of whether a second date can correct thin cloud.

Working rules. Change is anything that differs between dates, except lighting. Cloud in either image is seen through in preprocessing, so the detector never reads cloud as change.

REPORT

What can a flood-road map reliably tell users about roads hidden by cloud?

HANDLE

When either image is cloudy, can seeing through it with the other date find change as well as under clear sky?

Tested so far: thin cloud on the after image, corrected from the clear before image.

DETECT · THE TEST BED

Can we use synthetic clouds to represent real-life blockages in satellite imagery?

On hold until we can reliably see through real cloud.

Proposed method: detect, handle, report

GGS 680 · CLASSICAL

1 · Detect

Greyscale masks on each date: bright cores (our rule), the dark channel (He et al. 2011), and its rise from before to after (our rule).

GGS 662 · THE HANDLE GATE

2 · Handle, feature by feature

ASSESS	clear sky
CORRECT	thin cloud + clear second date
CANNOT ASSESS	opaque cloud

Correct = two-date de-hazing inside the cloud mask (our rule, after He et al. 2011), then the detector again (result 3). Filled pixels count as not observed.

WHAT THE MAP MAY SAY

3 · Report

Each statement about a hidden feature gets a tier:

RELIABLE

SYNTHETIC ONLY

RANGE ONLY

CHANGE DETECTOR

Random forest (Breiman 2001) on classical, texture (Haralick 1979) and DINOv2 (Oquab et al. 2024) cues: 79% on 798 roads in new areas (always “flooded”: 55.3%).

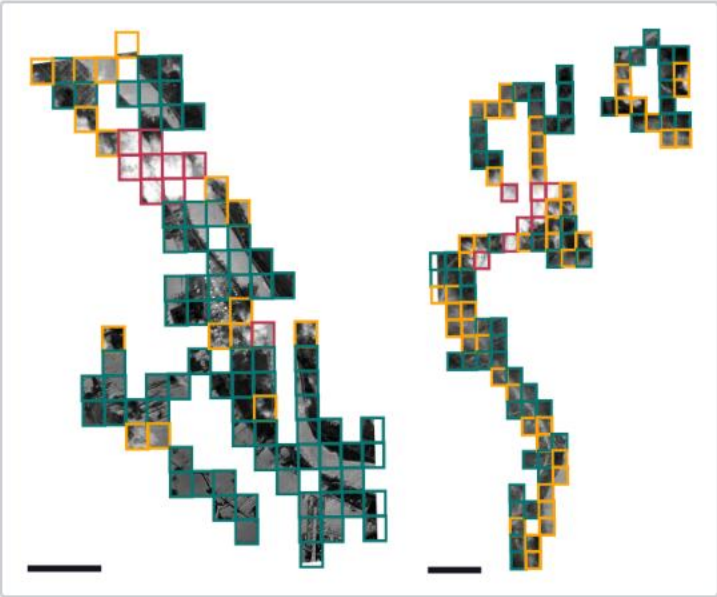
BASELINES AND CHECKS

Tile cloud rule; no de-hazing; He 2011. Folds blocked by 1.75 km area (Roberts et al. 2017), McNemar tests (McNemar 1947); a second event as the final test.

AGENTS PROPOSE, I DECIDE

Question, critic and validation agents draft and check; the handle gate is coded rules. Decided by hand: the problem, the questions, the cloud rule.

Study area and data

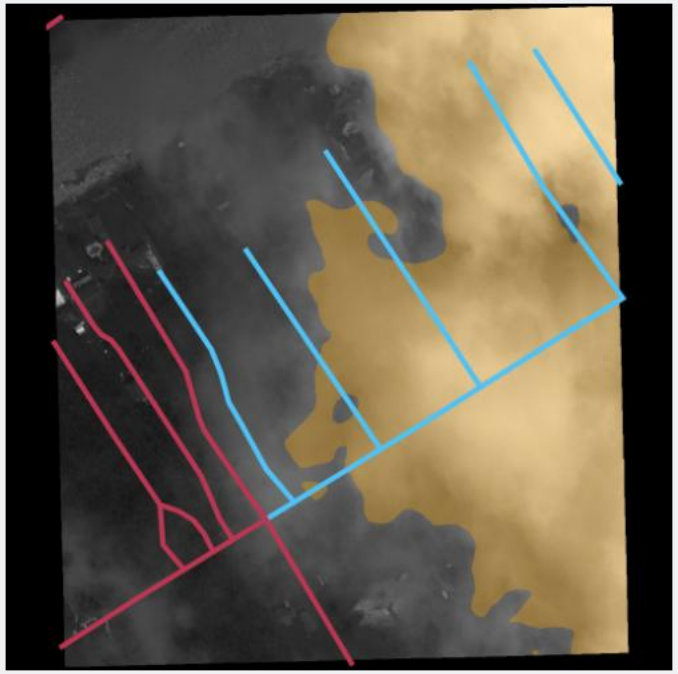


clear · thin cloud or haze · opaque

South (102 tiles) and west (115 tiles) after images, north up, bar 2 km; 29.7°N 90.0°W and 29.8°N 90.1°W · Imagery: Maxar via SpaceNet 8

Imagery	Maxar Open Data before and after Hurricane Ida (landfall 29 Aug 2021), 0.5 m; image dates unconfirmed; greyscale
Roads	1,037 OpenStreetMap roads with SpaceNet 8 flood labels (Hänsch et al. 2022), on 217 of the 226 development tiles
Split	186 train · 40 validation · 40 test tiles, plus a 333-tile north negative set; the test tiles, the negative set, Germany and Louisiana-West have never been scored
Cloud labels	HRC_WHU (Li et al. 2019): 150 Google Earth images, 0.5–15 m, expert cloud masks; 120 development, 30 sealed; academic use only
Unit	One road = its OSM line plus a ~5 m band each side; tiles are the fold unit; dates aligned to ~0.6 m
Quality issues	Flood labels are “yes” or missing, never “no” · 17 tiles under opaque cloud · one after image for all development tiles

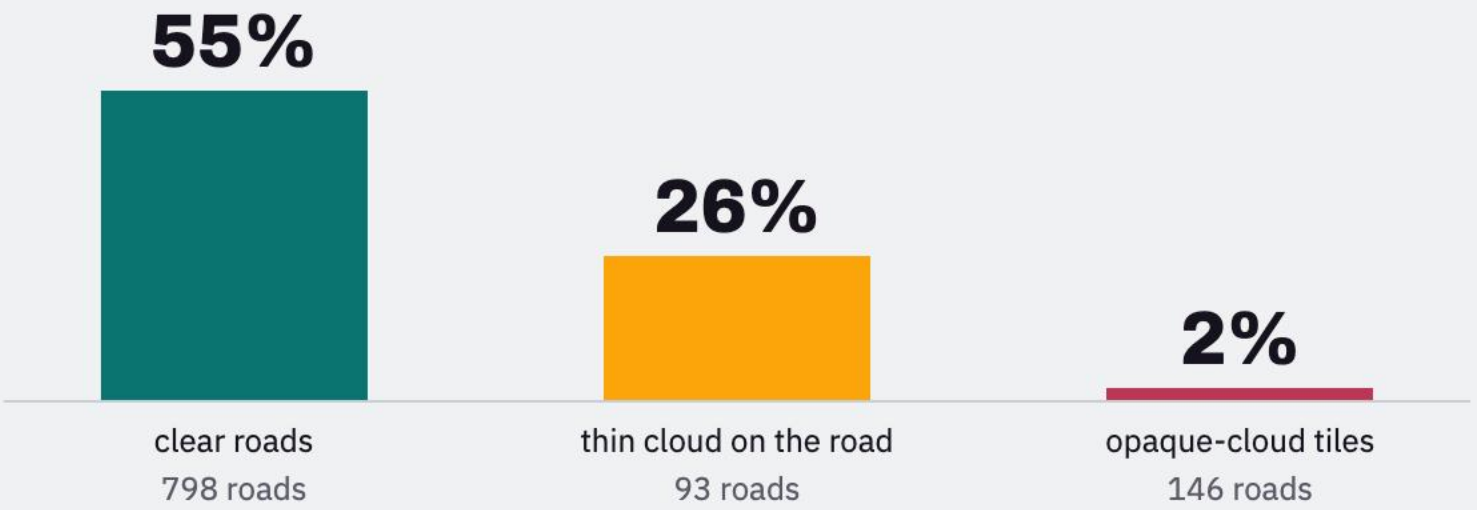
Under cloud, far fewer roads get a flood label



labelled “flooded” — thin cloud

no flood label (reads as dry in this dataset)

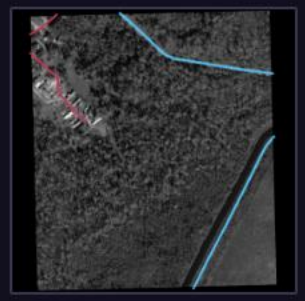
ROADS LABELLED “FLOODED”, BY CLOUD ON THE ROAD



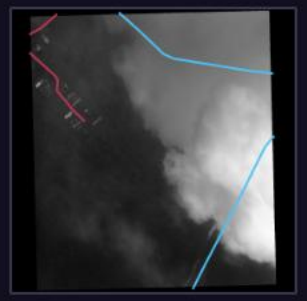
Analysts “only attribute what was visible” (Hänsch et al. 2022), so for this dataset a flooded road under cloud reads as dry.

Part of the gap is geography: the south scene is dry around its cloudy tiles. Left: an example tile, 2_15_45, after image · Imagery: Maxar via SpaceNet 8

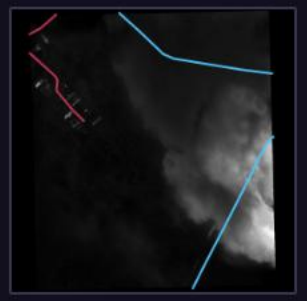
A second image lifts some thin cloud



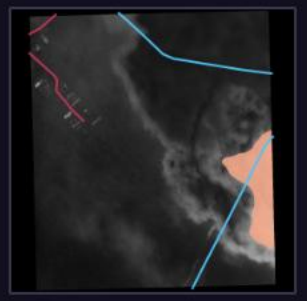
BEFORE · CLEAR



AFTER · CLOUD



HE 2011



TWO-DATE

Road picked by a fixed rule: the most metres under thin cloud (481 m, tile 2_23_44).

Red = labelled flooded · blue = no flood label · orange = opaque, cannot assess.

Haze model as used by He et al. (2011), whose de-hazing needs no before image: the baseline. Two-date de-hazing is our rule. Imagery: Maxar via SpaceNet 8.

SYNTHETIC HAZE · 94 CLEAR TILES

	none	He 2011	two-date
grey error (0–255)	57	27	37
detail kept (1 = all)	0.82	0.84	1.02
clear altered	0%	85%	<0.1%

He 2011: de-hazes without the before image.

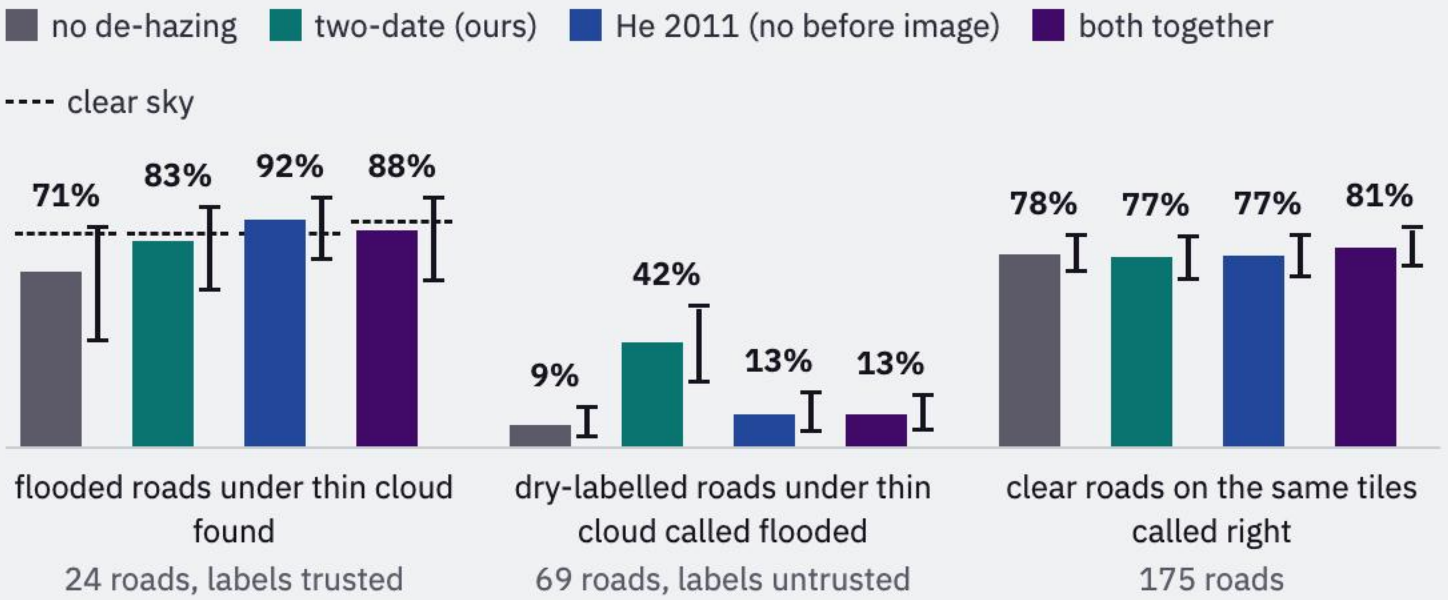
CAVEAT

The synthetic haze follows our model (favours it), and two-date leaves clear ground alone by design; faint haze outside the mask is left alone. Here, little road detail returns.

FLOOD CALLS · EXPLORATORY

Flooded roads under thin cloud found, model trained on de-hazed images: 20 of 24 two-date, 22 He 2011, 17 with none. Not significant: result 3.

De-hazing finds 3–5 more of 24 flooded roads, not significant



Change is always found by comparing the before and after images; only the de-hazing of the after image differs. Whiskers: 95% interval, tiles resampled (Efron 1979). Models trained on the image kind they are tested on (5 folds by tile). Both together: two-date and He cues in one model. Dashed: clear sky on the same tiles, 81 of 95 flooded roads found (86 with both together). Labels: SpaceNet 8 (Hänsch et al. 2022); imagery: Maxar via SpaceNet 8.

AGAINST NO DE-HAZING

Flooded found: two-date 4 fixed, 1 broken; He 5 fixed, 0 broken; clear roads unchanged. To show it counts: more trusted flooded roads, from a second event.

AGAINST TWO-DATE

Flooded found and clear roads alike. He and both together call fewer cloudy dry-labelled roads flooded (9 and 9 against 29, $p < 0.001$). Next: smooth the ring (He et al. 2013).

CLOUD AS AN EXTRA VARIABLE

No more flooded found (16 of 24 against 17); fewer cloudy dry-labelled roads called flooded (3 against 6), likely because cloudy roads were labelled dry.

Feasibility, limits and next steps

Feasible

- Data, masks, de-hazer and the held-out harness all run
- Cloud spotter (bright rule + texture + DINOv2): F1 0.87 against 0.65 for the dark channel (HRC_WHU)
- Label gap measured: 55%, 26%, 2% labelled flooded (partly geography)
- Each test runs in under an hour
- By semester end: a frozen see-through step for thin cloud, and report tiers

Limits

- One event, one sensor, roads only; all real cloud in one after image; cloud shadows untested
- Tile-split scores run about 4 points high; area-blocked folds from now on
- Two-date not yet better than He 2011 on flood calls
- De-hazing: only 24 trusted flooded roads, seen before; cloudy dry-labelled roads flagged

Next steps

- See through cloud on either date, as a preprocessing step
- Smooth the ring at the cloud edge
- Test the best de-hazing on a second event
- Tie it back into road detection and the flood model
- Report tiers and maps
- Later: synthetic clouds, once real cloud can be seen through

QUESTION FOR DISCUSSION

When a map cannot see the ground, should it say “cannot assess”, show a range, or fill it in from another date?

Scored in new areas: about 4 points lower, one tool still helps

Scored in new areas

78.6%

flood calls on 798 roads, folds blocked by 1.75 km area (mean of 21 fold seeds)

Split by tile it read 82.5%: 3.9 points high (95% CI +2.3 to +6.0), in 21 of 21 seeds. Neighbouring tiles share roads; smaller training sets explain only 0.25 points. Road finding is unaffected (F1 0.685 against 0.681).

The one tool that helps

+1.3

points from the DINOv2 change vector (95% CI +0.4 to +2.4), better in 21 of 21 seeds

Spotter gate and both de-hazers: within noise on all roads. He 2011 alone costs 1.1 points. Next: confirm the vector on more data before it joins the detector.

Benchmark · research only

83.0%

DINOv2 + DINOv3 cues and change vectors, against 79.2% for the detector (one fold seed)

+3.8 points (95% CI +1.2 to +6.7), 51 roads fixed, 21 broken, McNemar p = 0.0005. DINOv3 (Siméoni et al. 2025) is the bar for the open model, not part of it.

Spatially blocked cross-validation (Roberts et al. 2017): 63 blocks of 3 × 3 tiles, 5 folds, roads shared with a held-out block dropped from training. 226 development tiles; test and sealed data untouched. Labels: SpaceNet 8 (Hänsch et al. 2022).

Human review: what I accepted, changed or rejected

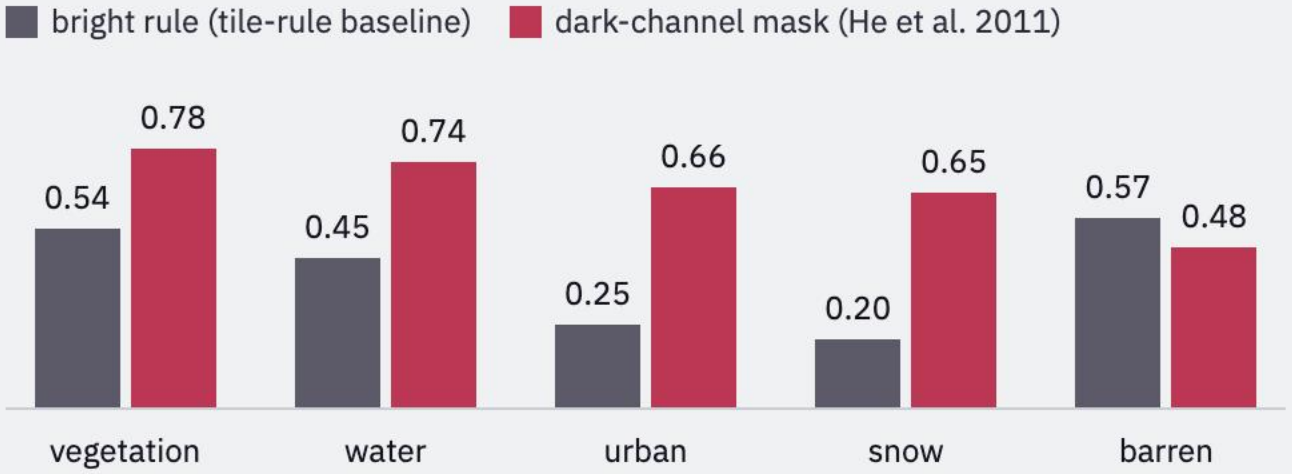
REJECTED	Handle question “the after image alone finds change”: reworded to seeing through cloud on either date first.	“how are you going to find change with the after image alone?”
CHANGED	Haze layer: a 2,500 m² minimum cloud size added.	“way too sensitive and catching houses as clouds”
CHANGED	Non-significant p-values: each is now an opportunity for further research.	“dont put a p-value that isnt significant … state what we will do to improve it”
ADDED	Ring at the cloud edge: five fixes tested, plus a literature check.	“a slight ring around the cloud that did not fully get removed”
ADDED	Cross-validation blocked by area: tile splits were about 4 points high.	“can you do a cross validation so that you dont overfit”
TESTED	Both de-hazers in one model (21 of 24 found); both masks add nothing.	“cant we use both He and two date?”
RULED	Sealed tiles stay sealed until testing time; synthetic clouds wait; change = anything but lighting; four critic changes adopted.	“Sealed tiles should stay sealed until testing time”

AI USE

Claude Code (Anthropic) agents drafted questions, critiques, code, figures and slides; I chose the problem, questions and cloud rule. All 38 decisions are logged, most in my words (11 changed, 7 rejected). Agent agreement is never counted as validation.

Cloud masks against human cloud labels

F1 AGAINST EXPERT CLOUD MASKS, BY LAND COVER



Bright rule: a pixel is cloud if it is very bright (grey above 200 of 255) and smooth. Usually right when it fires, but misses thin cloud.

Dark channel: the darkest pixel in each 20 m square. Clear ground nearly always has one (shadow, trees, water); under cloud even it turns bright. Finds more cloud, but bright bare ground can fool it.

0.65

F1, dark-channel mask (precision 0.62, recall 0.68)

0.41

F1, bright rule (precision 0.81, recall 0.28)

120 HRC_WHU development images, one date each (Li et al. 2019; 30 sealed).

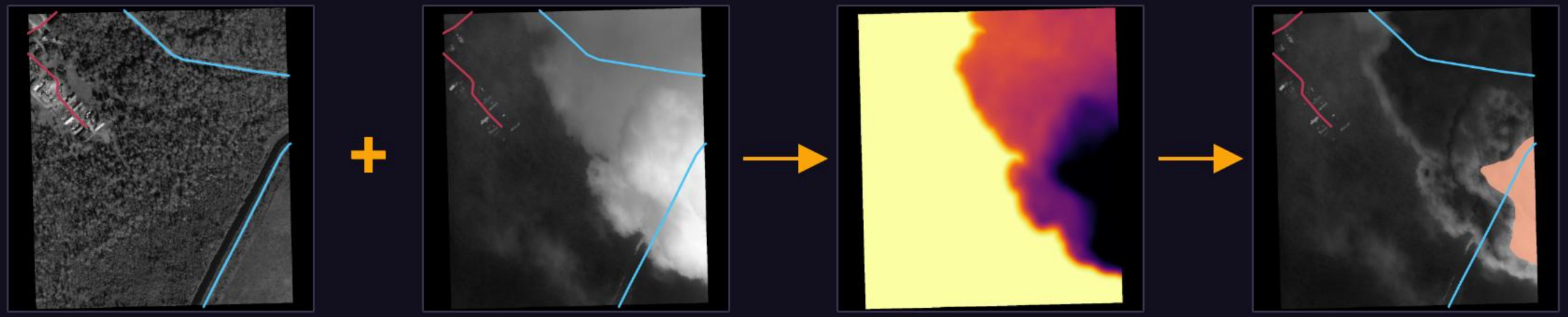
TRAINED SPOTTER · RESEARCH ONLY

Bright rule + texture (Haralick 1979) + DINOv2: F1 0.87, +0.22 (95% CI +0.19 to +0.25), better on 112 of 120 images.

MASK-BIAS CHECK

Synthetic haze on 769 clear roads: under thin haze the mask flags flooded and dry roads alike.

How the two images are layered



1 · BEFORE

Clear reference: the ground's darkest pixels

2 · AFTER

Ground dimmed by t, plus airlight A(1 – t)

3 · HAZE t

Inside the cloud mask only; houses stay at 1

4 · CORRECTED

Orange: opaque (t below 0.2), cannot assess

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t = (A − dark after) ÷ (A − tone(dark before)) · corrected = (after − A) ÷ t + A
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Standard haze model and dark channel as used by He et al. (2011). Our rules: the before image's darkest pixels in place of He's zero assumption, the tone curve, a 2,500 m² cloud minimum, a 20 m fade past the mask, and t below 0.2 left as opaque. Imagery: Maxar via SpaceNet 8, tile 2_23_44.

References and code

CITED ON THESE SLIDES

Li et al. 2019	HRC_WHU cloud masks · ISPRS J. Photogramm. Remote Sens.
He, Sun & Tang 2011, 2013	Dark channel; guided filter · IEEE TPAMI
Oquab 2024; Siméoni 2025	DINOv2 (TMLR); DINOv3 benchmark (arXiv) · Meta AI
Haralick 1979	Texture features · Proc. IEEE
Roberts et al. 2017	Blocked cross-validation · Ecography

BACKGROUND

Canty & Nielsen 2008 (Remote Sensing of Environment) · Portalés-Julià et al. 2023 (Scientific Reports) · Qu, Li & Huang 2026 (Remote Sensing of Environment) · Shen et al. 2015 (IEEE Geoscience and Remote Sensing Magazine) · Bruzzone & Bovolo 2013 (Proc. IEEE)

METHOD ATTRIBUTIONS

Breiman 2001, random forests (Machine Learning) · McNemar 1947, paired proportions (Psychometrika) · Efron 1979, the bootstrap (Annals of Statistics)

Data: SpaceNet 8 labels, Hänsch et al. 2022 (licences differ) · HRC_WHU (academic use only)

CODE

github.com/Jrog378/GeoAI-Roads (private; access on request): masks, de-hazing, notebooks

Full references: cited

He, K., Sun, J. & Tang, X. (2011). Single image haze removal using dark channel prior. *IEEE TPAMI* 33(12), 2341–2353. doi:10.1109/TPAMI.2010.168

He, K., Sun, J. & Tang, X. (2013). Guided image filtering. *IEEE TPAMI* 35(6), 1397–1409. doi:10.1109/TPAMI.2012.213

Li, Z., Shen, H., Cheng, Q., et al. (2019). Deep learning based cloud detection for medium and high resolution remote sensing images of different sensors. *ISPRS J. Photogramm. Remote Sens.* 150, 197–212. doi:10.1016/j.isprsjprs.2019.02.017

Oquab, M., Darcet, T., Moutakanni, T., et al. (2024). DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*. openreview.net/forum?id=a68SUt6zFt

Siméoni, O., Vo, H. V., Seitzer, M., et al. (2025). DINOv3. *arXiv* 2508.10104 (Meta AI). doi:10.48550/arXiv.2508.10104

Haralick, R. M. (1979). Statistical and structural approaches to texture. *Proc. IEEE* 67(5), 786–804. doi:10.1109/PROC.1979.11328

Roberts, D. R., Bahn, V., Ciuti, S., et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40(8), 913–929. doi:10.1111/ecog.02881

Hänsch, R., Arndt, J., Lunga, D., et al. (2022). SpaceNet 8 – The detection of flooded roads and buildings. *CVPR Workshops*, 1471–1479. doi:10.1109/CVPRW56347.2022.00153

Breiman, L. (2001). Random forests. *Machine Learning* 45(1), 5–32. doi:10.1023/A:1010933404324

McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* 12(2), 153–157. doi:10.1007/BF02295996

Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *Annals of Statistics* 7(1), 1–26. doi:10.1214/aos/1176344552

Checked against Crossref, TMLR and the CVF open-access copy (evidence 08). Breiman, McNemar and Efron are method attributions.

Full references: background

Canty, M. J. & Nielsen, A. A. (2008). Automatic radiometric normalization of multitemporal satellite imagery with the iteratively re-weighted MAD transformation. *Remote Sens. Environ.* 112(3), 1025–1036. doi:10.1016/j.rse.2007.07.013

Portalés-Julià, E., Mateo-García, G., Purcell, C., et al. (2023). Global flood extent segmentation in optical satellite images. *Scientific Reports* 13, 20316. doi:10.1038/s41598-023-47595-7

Qu, Y., Li, J. & Huang, X. (2026). AnyTime-CD: Self-supervised change detection in arbitrary-length dense time series of high-resolution remote sensing images. *Remote Sens. Environ.* 334, 115239. doi:10.1016/j.rse.2026.115239

Shen, H., Li, X., Cheng, Q., et al. (2015). Missing information reconstruction of remote sensing data: A technical review. *IEEE Geosci. Remote Sens. Mag.* 3(3), 61–85. doi:10.1109/MGRS.2015.2441912

Bruzzone, L. & Bovolo, F. (2013). A novel framework for the design of change-detection systems for very-high-resolution remote sensing images. *Proc. IEEE* 101(3), 609–630. doi:10.1109/JPROC.2012.2197169

Data: SpaceNet 8 imagery from the Maxar Open Data Program; the paper states a Creative Commons 4.0 non-commercial licence, the SpaceNet site CC BY-SA 4.0. HRC_WHU: academic use only.